One cache entry changes

First coordinate update: $\tau = 1/7$ and $T = 1/7$.

Entry	Before	After selecting 0
Gain B	$(0, 0)^T$	$(4/7, 0)^T$
Residual G_0	$(4, 0)^T$	$(4/7, 0)^T$
Energy e_0	16	16/49
Residual G_1	$(0, 1)^T$	$(0, 1)^T$
Energy e_1	1	1
Total	17	65/49

Exact cache state transition. Initialization computes the full residual once.